

Označme si vektory v $\text{Span}\{1, x, x^2\}$ jako $\text{Span}\{\vec{u}_1, \vec{u}_2, \vec{u}_3\}$

Budeme hledat ON bázi $B = (\vec{u}_1, \vec{u}_2, \vec{u}_3)$ přes

Gram-Schmidtovu ortogonalizaci.

$$\vec{u}_1 = \frac{1}{\sqrt{\int_1^5 1^2 dx}} = \frac{1}{2}$$

$$\vec{u}_2 = \frac{v_1 - w_1}{\|v_1 - w_1\|} \quad w_1 = \left\langle \frac{1}{2}x, x \right\rangle \cdot \frac{1}{2} \quad w_1 = 3$$

$$\vec{u}_2 = \frac{x-3}{\|x-3\|} = \frac{x-3}{\sqrt{\int_1^5 (x-3)^2 dx}} = \frac{\sqrt{3}}{4} \cdot (x-3)$$

$$\begin{aligned} \vec{u}_3 &= \frac{v_2 - w_2}{\|v_2 - w_2\|} & w_2 &= \left\langle \frac{1}{2}x^2, x^2 \right\rangle \cdot \frac{1}{2} + \left\langle \frac{\sqrt{3}}{4}(x-3), x^2 \right\rangle \cdot \frac{\sqrt{3}}{4}(x-3) \\ & & &= \frac{31}{3} + 6(x-3) = \frac{31}{3} + 6x - 18 \\ & & &= -\frac{23}{3} + 6x \end{aligned}$$

$$\vec{u}_3 = \frac{x^2 - 6x + \frac{23}{3}}{\|x^2 - 6x + \frac{23}{3}\|} = \frac{x^2 - 6x + \frac{23}{3}}{\sqrt{\int_1^5 (x^2 - 6x + \frac{23}{3})^2 dx}} = \frac{3\sqrt{5}}{16} \cdot \left(x^2 - 6x + \frac{23}{3} \right)$$

$$B = \left\{ \frac{1}{2}, \frac{\sqrt{3}}{4}(x-3), \frac{3\sqrt{5}}{16} \left(x^2 - 6x + \frac{23}{3} \right) \right\}$$

b) $\sin x$

$$P(\sin x) = \left\langle \frac{1}{2}, \sin x \right\rangle \cdot \frac{1}{2}$$

$$+ \left\langle \frac{\sqrt{3}}{4} (x-3), \sin x \right\rangle \cdot \frac{\sqrt{3}}{4} (x-3)$$

$$+ \left\langle \frac{3\sqrt{5}}{76} \left(x^2 - 6x + \frac{23}{3} \right), \sin x \right\rangle \cdot \frac{3\sqrt{5}}{76} \left(x^2 - 6x + \frac{23}{3} \right)$$

(poveljen izračun ve Wolfram alpha)

$$= \frac{15}{128} \left(x^2 - 6x + \frac{23}{3} \right) (6(\sin(1) + \sin(5)) + \cos(1) - \cos(5))$$

$$+ \frac{3}{76} (x-3) (-\sin(1) + \sin(5) - 2\cos(1) - 2\cos(5))$$

$$+ \frac{1}{4} (\cos(1) - \cos(5))$$

$$= \frac{15}{128} (\cos(1) - \cos(5) + 6\sin(1) + 6\sin(5)) x^2$$

$$- \frac{1}{64} (69\cos(1) - 21\cos(5) + 282\sin(1) + 258\sin(5)) x$$

$$+ \frac{1}{128} (297\cos(1) - 3\cos(5) + 762\sin(1) + 618\sin(5))$$

po aproksimaciji sinu a cosinu

$$\approx -0,05251x^2 - 0,33751x + 1,60127$$

$$c) \quad \left\| \sin(x) - (-0,05257x^2 - 0,33157x + 1,602127) \right\|$$

$$= \int_0^1 (\sin(x) + 0,05257x^2 + 0,33157x - 1,602127)^2 dx$$

$$= \underline{\underline{0,318794}}$$